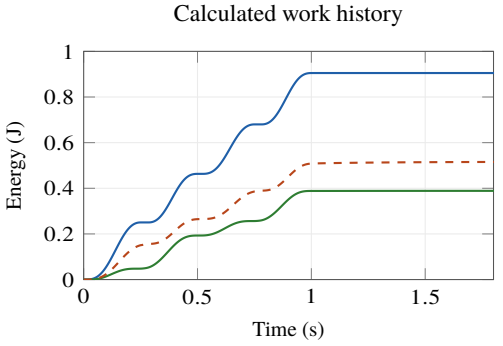
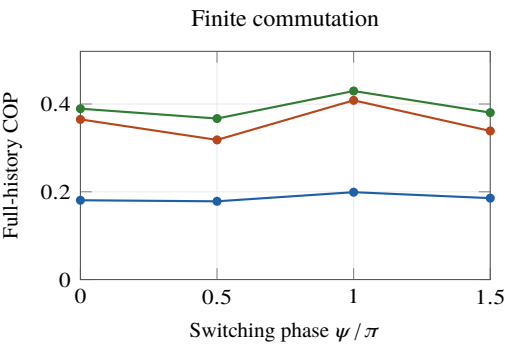

Calculated work history
Energy (J)
1
0.8
0.6
0.4
0.2
0
0
0.5
1
1.5
Time (s)
Gross input D
Useful output W_R
Return + other heat


Finite commutation
Full-history COP
0.4
0.2
0
0
0.5
1
1.5
Switching phase ψ / π
R = 1 Ω
R = 4 Ω
R = 16 Ω